

The Fall of the Florida Panther... and the Rise?

New York City, Miami, Los Angeles, Boston, and Chicago are notable metropolises around the U.S. These cities are densely populated and boast many forms of entertainment and food that attract people from around the world. Yet, these cities came at a cost due to urbanization and its effects have been catastrophic to animals. Urbanization facilitated the loss of natural habitats not only threatens species like tigers, orangutans, Indian elephants, whales, and mountain gorillas, but also the survival of the Florida panther. Just like the other species previously mentioned, the Florida panther faces endangerment due to habitat loss, poaching, and the challenge of adapting to rapidly changing environments. In this urban landscape, Florida's expansion exacerbates the challenges facing wildlife conservation. With the state experiencing rapid population growth and urban sprawl, habitat loss intensifies, placing additional pressure on species like the Florida panther. The increasing development encroaches upon crucial panther habitats, further threatening their survival. Thus, amidst the vibrant tapestry of city life, the urgency to protect and preserve natural habitats becomes even more pronounced. To counteract these effects and increase the Florida panther population, the government must prioritize habitat conservation, anti-poaching measures, and aiding adaptation to urban environments, recognizing the critical intersection between urbanization and wildlife preservation.

The increased urbanization has led to more construction. The increase in construction occurs in unpopulated land which facilitates tourism, jobs, and migration. According to a study by the Florida Fish and Wildlife Conservation Commission (FWC), habitat fragmentation caused

by urban development has significantly reduced the available habitat for Florida panthers (Florida Fish & Wildlife Conservation Commission, p.1). The fragmentation disrupts panther movement patterns, isolating populations and hindering genetic diversity. This lack of genetic variation increases the susceptibility of panthers to diseases and reduces their ability to adapt to changing environmental conditions. Due to habitat fragmentation, there is a serious threat to the Florida panther population by limiting their ability to roam and find suitable mates. The lack of mating decreases offspring and as a result, there is a higher risk of inbreeding depression and decreased population viability over time. The decrease of population variability makes it difficult for panthers to adapt to civilization. Research conducted by the National Park Service (NPS) highlights the importance of natural corridors for maintaining genetic connectivity among panther populations (National Park Service, p.1). However, the expansion of roads and urban areas in Florida has led to the destruction and fragmentation of these corridors, further isolating panther populations, and hindering their ability to disperse and establish new territories. The loss of natural corridors exacerbates the challenges faced by Florida panthers in finding mates and establishing viable populations. Without these corridors, panthers face increased risks of inbreeding and reduced genetic diversity, ultimately jeopardizing their long-term survival.

To combat habitat fragmentation, the establishment of protected areas and conservation initiatives, such as the Florida Panther National Wildlife Refuge and the Panther Posse program (Florida Gulf Coast University, p.1), has shown promising results in mitigating habitat loss and protecting panther habitats. These efforts focus on acquiring and preserving critical habitat areas, restoring degraded habitats, and implementing land management practices that promote panther conservation (Florida Gulf Coast University, p.1). These programs also educate people surrounding panthers which helps bring awareness to their endangerment. Investing in protected

areas and conservation programs is essential for safeguarding the remaining habitats of Florida panthers. By preserving key areas and implementing habitat restoration measures, we can enhance the resilience of panther populations and provide them with the necessary resources for survival and reproduction.

Another issue that is responsible for the panther population decline is poaching. The illegal wildlife trade poses a significant threat to Florida panthers, with poachers targeting them for their pelts and body parts. In 1832, Florida rewarded panther bounty hunters as and so this led to the decline of the panther population (Gutknecht, 2020). Poaching continues to be a major factor driving the decline of panther populations (McKenney. M, McKenney. D, 2023), despite legal protections in place. Poaching not only reduces panther numbers directly but also disrupts social structures and breeding dynamics within populations. Strengthening law enforcement efforts and implementing stricter penalties for poaching are crucial steps in combatting this illegal activity and protecting panther populations. Encounters between panthers and humans often result in conflicts, particularly in areas where urban development encroaches upon panther habitats. Implementing measures to mitigate human-wildlife conflicts, such as the use of predator-proof fencing, livestock guardian animals, and community education programs, can help reduce retaliatory killings and promote coexistence between humans and panthers. To counter poaching, fines should increase for intentionally killing endangered species. In 2015, the Center for Biological Diversity offered a \$15,000 reward for information regarding an intentional killing of a panther (Center for Biological Diversity, 2015). The Endangered Species Act also states that harming a Florida Panther can rise to \$100,000 per person and a year in prison (Center for Biological Diversity, 2015). There should continue to be an increase in fines and prison sentences for endangered species by improving enforcement of anti-poaching laws for

panthers and expanding monitoring resources could also reduce intentional killings. Using updated technologies like camera traps, DNA analysis, and wildlife forensics can help identify offenders. Combining these methods would see an increase in the panther population.

Furthermore, to maintain habitat connectivity within urbanized landscapes is essential for facilitating panther movement and dispersal. For instance, the U.S Fish and Wildlife Service has emphasized the importance of green corridors and wildlife crossings in urban areas for enabling panthers to traverse fragmented habitats safely (Bodine, 2022). Incorporating wildlife-friendly infrastructure, such as wildlife crossings and greenways, into urban planning initiatives can enhance habitat connectivity and reduce the barriers to panther movement imposed by urban development. Encouraging human-wildlife coexistence through measures such as responsible pet ownership, secure garbage disposal, and habitat-friendly landscaping can help mitigate conflicts between panthers and residents in urban areas. Educational outreach programs conducted by organizations like the FWC aim to raise awareness about panther behavior and promote tolerance towards these animals. Promoting coexistence between humans and panthers is essential for ensuring the long-term survival of panther populations in urbanized environments. By fostering understanding and respect for panthers among residents, we can reduce negative interactions and create a more harmonious relationship between people and wildlife.

With an increase of migration and urbanization, this leads to more drivers on the road. Florida, already known for its frequent speeders, poses a danger to these panthers. Fort Myers News-Press quoted Jay Anderson, an executive director of Stay Alive.....Just Drive! Inc., a safe driving education program based in Fort Myers who stated: “Once again driver error, speeding, impairment, and the lack of restraints prove to be deadly” (Braun, p.1). The lack of driver

awareness provides an unsafe environment for these panthers, especially if there are zero panther crossing signs. The lack of panther crossing signs influences drivers to speed as drivers slow down when there are animal crossing signs. The Florida Fish and Wildlife Conservation Commission stated that “Panther deaths caused by vehicle collisions have been sharply reduced in areas where crossings and fencing are in place” (Florida Fish & Wildlife Conservation Commission, p.1). Because of the number of panthers dying from speeding, there should be an increase in fines for speeding. The average fine for speeding in a nighttime panther zone is around \$200 (Florida Fish & Wildlife Conservation, p.2). Increasing Higher fines act as an effective as raising speeding fines leads more drivers to comply with posted speed limits. This could reduce collisions with panthers crossing roads. When people are driving over the speed limit, there is less time to stop or brake in time when a panther crosses unexpectedly. Higher speeds also increase impact forces which would increase the probability of a panther dying if hit. This habitat fragmentation from road mortality threatens their existence. The added revenue from larger fines could be used by local authorities specifically to install more panther crossing signs, lighting, fencing, and wildlife bridges to mitigate road dangers. Steeper fines for speeding in designated panther zones also remind drivers that excess speed poses a real threat that contributes to their endangerment. To maintain the panther species, increasing speeding penalties incentivize slower driving, reduce direct mortality, provide funding for additional measures, and promote public consciousness which are important to facilitate and promote safety for panthers.

Alongside increasing speeding fines, increasing the panther crossings will allow them to adapt to the increased construction and drivers on the road. The Florida Department of Transportation is planning to allocate \$11.9 million to construct wildlife overpass and crossings (Thalheimer, I., & Gallardo, 2023). With this money, panthers will be able to migrate more

freely and safely. This is a positive for both people and wildlife because this increases population and reduces crashes against wildlife. While panthers are endangered species, this would benefit other animals such as bears, deer, wild hogs, manatees, and other animals who are affected by the expansion of roads. This maintains genetic connections between sub-populations. Crossings enable seasonal and dispersal movements for smaller animals like bobcats, foxes, and raccoons by providing safe passage across roads. In general, creating more wildlife crossings saves many species from vehicle impacts as they move around fragmented habitats.

However, some may agree that urbanization is necessary as it facilitates new jobs and helps the overall economy more. With many corporations such as Disney and Citadel moving their headquarters to Florida, many workers have also migrated. Mass migration has soared rent prices as there is a demand for housing. There is an increase in construction, notable in Southwest Florida as hotels, housing, beach resorts, and hospitals (Dorsey, p.1). Construction has increased because of Hurricane Ian which reduced tourism in Southwest Florida, decreasing economic power in the area. For this reason, there is an argument that supports the reallocation of money towards construction projects instead of environmental sustainability plans to lift economic instability from the area. In addition, there is a shortage of teachers as they are not compensated. Wink News reported that teachers work multiple jobs to afford housing and rejecting the Lee County proposal of \$32.8 million split across all public-school teachers (Moore, 2023). Also, wildlife crossings tend to be the largest expense when exploring measures to aid the panther population and as previously mentioned are going to cost above \$11.9 million (Thalheimer, I., & Gallardo, 2023). With the budget in some counties being less than others it would seem appropriate to allocate the money in other sectors such as urbanization planning and education system. While these arguments hold merit, humans are responsible for the population

decline of panthers. The extinction of a species in an ecosystem has negative impacts such as overpopulation since a predator will be eliminated, in this case the panther. With the invasive species population in python increasing, it is important to combat against them by preserving and aiding panthers and alligators. Controlling the population of native species in the ecosystem is important to have a stable environment that promotes biological diversity.

While humans are the major cause for the population decline in Florida panthers and other species, it's important to acknowledge this issue and act before they go extinct. The juxtaposition of bustling urban centers and dwindling wildlife populations underscores the urgent need for concerted action to mitigate the adverse effects of human activity on the environment. While major cities offering unparalleled opportunities for entertainment and commerce, their expansion has come at the expense of precious natural habitats and the survival of countless species, including the iconic Florida panther. As Florida's rapid development continues to encroach upon vital wildlife habitats, the imperative to prioritize conservation efforts becomes ever more pressing. By acknowledging our role in driving species decline and committing to proactive measures, such as habitat preservation, anti-poaching & speeding initiatives, and promoting coexistence with urban wildlife, we can strive to reverse the trend of biodiversity loss and ensure a sustainable future for both humans and wildlife alike. A construction project can be postponed, but the extinction of a species is irreversible.

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